

Designing for Transparency: Gaze-Augmented Collaborative Action Recognition with Vision-Language Models

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Abstract—Understanding how students collaborate in STEM learning activities is key to building scientific literacy; however, scaling observational studies remains a challenge for educational research. This paper investigates the use of vision-language models (VLMs) with zero-shot Chain-of-Thought (CoT) prompting for multimodal learning analytics (MMLA). We present a video-frame analysis pipeline that integrates gaze and hand detection with structured prompts to classify five collaborative actions: Communicating, Observing, Manipulating, Idle-OffTask, and Unclear. The pipeline mirrors human coding through rule-based definitions and produces transparent reasoning traces, enabling direct comparison with expert annotations. Data from two pilot sessions were coded by researchers ($\kappa = 0.73\text{--}0.84$). Multiple VLMs (*GLM-4.5V*, *InternVL-3.5*, *Gemini-2.5-Pro*, *GPT-5*) were then evaluated against this standard. Results demonstrate the feasibility of the pipeline and that CoT prompting can yield benefits in accuracy. More importantly, the approach provides transparency akin to human coding. Class-wise analysis shows that *Manipulating* and *Observing* actions were consistently well recognized, while *Communicating* proved more challenging. We discuss reliability, discrepancies between human and machine coders, the differences in how the models perform, and the potential of CoT-based pipelines to deliver explainable and scalable MMLA across diverse learning contexts.

Index Terms—Multimodal Learning Analytics, Group Work, Vision Language Model, Gaze Tracking, Smart Learning Environments

I. INTRODUCTION

Increasing student engagement in science is a key goal for society to build scientific literacy, critical thinking, and problem-solving skills essential for informed citizenship and navigating global challenges [1]. Understanding how students work together in collaborative science activities is a complex endeavor that has generally been researched by collecting rich data on small cohorts of students. This approach has its drawbacks since it is difficult to scale [2]. Recent advances in AI and multimodal learning analytics (MMLA), particularly with the introduction of vision large language models (VLMs), may provide new tools and methods for gaining insights into how students meaningfully interact with one another through the analysis, allowing researchers to scale up observations. VLMs utilize generative AI to take image and text inputs and generate text outputs. Therefore, a system can be created that sends video frames with instructions (prompts) to the VLM and then returns a textual response, which can be used to analyze the learning activity in a manner similar to a researcher observing the learning process. This approach would allow larger-scale observations of learning activities. However, the

challenges of implementing a transparent process and system grounded in various educational theories can be applied across multiple contexts, which provide reliable observations that need to be investigated. We have employed a Zero-Shot Chain-of-Thought (CoT), a prompting technique for Vision Language Models (VLMs) that involves incorporating the step-by-step reasoning required to arrive at the correct answer without explicit examples [3]. This approach guides the model to perform complex, multi-step reasoning tasks by showing it how to break down problems, rather than just presenting the solution [4]. Motivated by these considerations, this paper is guided by the following research questions:

- 1) To what extent can a VLM-based pipeline, augmented with gaze data, automate collaborative action recognition with human-interpretable reasoning?
- 2) What are the consistencies and discrepancies between VLM-generated reasoning traces and human expert coding in collaborative settings?

The paper investigates the process of extending an ongoing MMLA [5], [6] system with Zero-Shot-Chain-of-Thought to provide a method for automatic physical action recognition from video frames, offering insights into how students interact, specifically when collaborating in small groups. These insights are grounded through the lenses of educational theories and need to be applied by researchers to different constructs that align with human observations. An additional area for discussion is to examine how different VLMs perform in our system in comparison to human observers.

II. RELATED WORK

Many investigations into student behavior rely on human observers entering the learning situation and tracking the actions of the students, guided by a theoretical lens through which the results are interpreted. This has yielded excellent results and helped shape our understanding of teaching and how it is conducted [7]. However, this process is both time-consuming and costly, as it relies for the most part on human labor; consequently, many such studies use a small cohort from which detailed data/observations are gathered. The issue with this is collecting and analysing enough data to make descriptions and inferences at the desired level of granularity and fidelity. One of the goals of research into Multimodal Learning Analytics (MMLA) is to enable researchers to collect and analyse rich data from a larger cohort [8]. In the evolution of the field of MMLA, there has been a call for

more theory-driven development, relying on the education theory developed in traditional studies of learning [9]. There have been various approaches to theory-driven development, such as the work of Martinez-Maldonado, R. et al. [10] and Yan, L. et al. [11], which convert the MMLA data stream into human-understandable markers for analysis. Researchers have been investigating how to detect and interpret human behavioral interaction in learning environments, focusing on how computer vision and audio analysis can be used in conjunction with artificial intelligence, specifically different types of VLMs [12]. Other researchers have utilized gaze and hand motion [13] to support computer vision approaches and gain insights into collaborative activities. Yu et al. [14] have presented an end-to-end framework that focuses on temporal action detection using skeletal data, aiming to provide action pedagogical reports for teachers. Whitehead et al. [15] developed a data pipeline that included feature extraction focused on socially shared regulation of learning, with an inter-reliability process involving the random selection of frames (2.5% of the dataset) by both human coders and the pipeline, achieving high reliability.

There is an opportunity to develop more explainable, transparent, and robust AI pipelines for education [16].

III. METHODOLOGY

A. Context

The data was collected from two pilot sessions in Europe and Asia, where a pair (researchers and university students) was tasked to use a Micro:Bit kit that teaches programming and electronics with a breadboard and components¹. An activity was developed by the authors that included a small intro video and then the task was to build a simple state machine with a red, yellow, and green LED triggered by buttons. This pilot is part of a larger joint project to develop an MMLA system that captures different types of audio, spatial information, and human interaction [5], [6].

B. Collaborative Problem-Solving Actions

From the perspective of defining collaboration as working together towards a common goal, communication, cooperation, and responsiveness constitute core elements of students' problem-solving skills in digital learning environments [17]. Working from the Graesser et al. [18] and current research in collaborative problem-solving frameworks used in MMLA [19]–[21], we developed an action coding scheme consisting of five mutually exclusive categories: communicating, working, observing, distracted, and unclear, as detailed in Table I. Human coders followed the descriptive definitions when labeling the frames, whereas machines adopted a rule-based formalization of the same definitions. The latter were specified in the VLM prompts using explicit conditions on gaze and hand status (e.g., GAZE directed at a work-related element AND HAND contacting the same element → Manipulating).

¹<https://kitronik.co.uk/collections/microbit-and-accessories/products/inventors-kit-for-the-bbc-micro-bit>

C. VLM pipeline

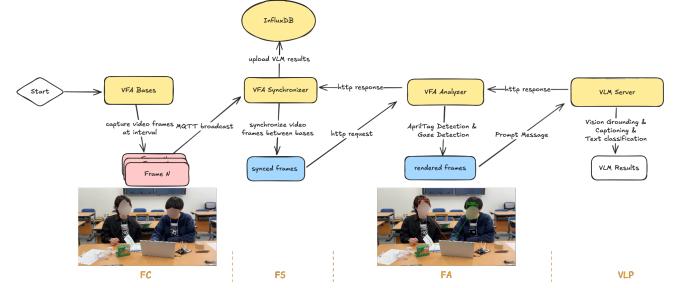


Fig. 1. Proposed end-to-end video frame analysis(VFA) pipeline

The primary objective of this paper is to provide an end-to-end frame analysis tool that automates the process of behavior analysis on actions. The proposed pipeline shown in Fig. 1 consists of four components that enable the acquisition, synchronization, analysis, and semantic interpretation of video data streams.

Frame Capture (FC): The VFA bases capture video frames from the video stream at a fixed interval, and then broadcast the metadata (camera angle description and frame save path) to the VFA synchronizer via MQTT.

Frame Synchronization (FS): The VFA synchronizer listens on the corresponding MQTT channel and temporally aligns the messages across bases. It generates a unified set of time-synchronized frames from multi-angle bases and then ensembles the frames with the context of participant and angle descriptions to the analyzer via HTTP request.

Frame Analysis (FA): The VFA analyzer provides low-level pre-processing on the synchronized frames, including AprilTag [22] and Gaze detection [23], and renders the frames with overlaid annotations (bounded ID, gaze vectors). Then, the analyzer constructs the structured prompt message and transmits it to the VLM server for higher-level multimodal reasoning.

Vision-Language Processing (VLP): The VLM server (deployed locally or in the cloud) applies vision grounding, captioning, and text classification to the prompt message (rendered frames plus textual context). The resulting semantic representations are transmitted to the VFA analyzer & synchronizer and stored in InfluxDB for archival and later retrieval.

D. Prompt Engineering

To enable consistent and transparent machine coding, we developed a structural prompt engineering pipeline (Fig. 2). The design mirrors the human annotation process but translates it into a stepwise reasoning framework for the VLM: (1) *Persona assignment* – assigning the VLM an expert analyst role; (2) *Context specification* – providing camera setup and participant descriptions; (3) *Grounding* – identifying participants via overlaid AprilTag IDs; (4) *Captioning* – describing each person's gaze, hands, position, and clothing; (5) *Classifying* – applying rule-based if-else reasoning on gaze and hand status; (6) *Formulating* – structuring the response in validated JSON format.

TABLE I
CODING SCHEME OF COLLABORATIVE ACTIONS WITH GAZE AND HAND CONTEXT

Category	Definition
Communicating	Person is actively communicating with others. This happens when: (1) Looking at work items (screen, documents, hardware) while clearly pointing at those items, OR (2) Looking at another person while gesturing or pointing with hands.
Observing	Person is watching or monitoring but not actively communicating or manipulating. This happens when: Looking at work items or people, but hands are either resting/inactive, hovering without touching, or touching something different from what they're looking at.
Manipulating	Person is directly working with something. This happens when: Looking at work items (screen, keyboard, documents, hardware) while at least one hand is clearly touching or contacting the same item they're looking at.
Idle-OffTask	Person is not engaged in the work task. This happens when: Looking at personal items (phone, snacks) or looking outside the camera frame. Hands can be doing anything.
Unclear	Cannot tell what the person is doing. This happens when: Cannot clearly see where they're looking or what their hands are doing due to blocking, blur, or poor visibility.

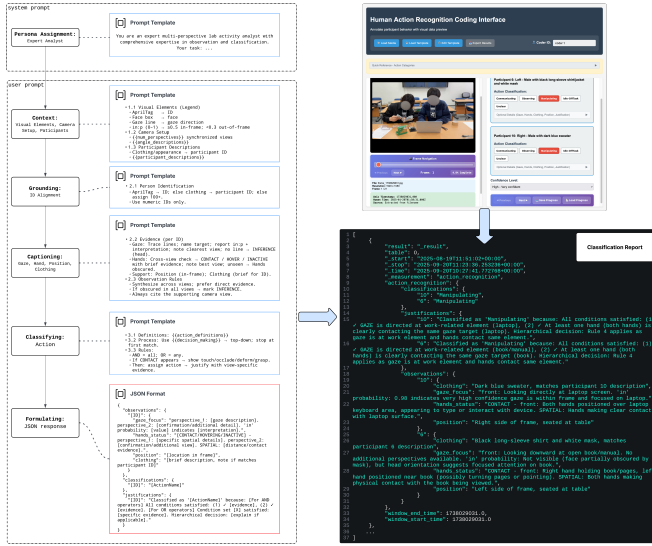


Fig. 2. CoT prompt template & Human coding interface

E. Perspective of Ground Truth

Three researchers coded both sessions using the Human Action Recognition Coding Interface (see Figure 2), which provides the coding scheme as buttons with additional fields for details. Video frames were captured from a single camera. The coding scheme was agreed upon by the researchers and served as the rules for the reasoning chain guiding the prompt engineering, aiming to minimize hallucinations. The inter-rater reliability (Cohen’s κ) was: Location 1, P1 $\kappa = 0.73$, P2 $\kappa = 0.78$; Location 2, P3 $\kappa = 0.84$, P4 $\kappa = 0.78$ —demonstrating substantial to high agreement sufficient to form the ground truth.

IV. EXPERIMENTS

Video frames were captured in a single-front view from a Logitech C920 camera every 30 seconds, yielding 110 frames in total (session A: 89 frames without occlusion; session B: 21 frames with frequent occlusion). Each frame may contain zero, one, or two participants, resulting in a maximum of 220 person-frame coding instances. After applying the majority-

consensus procedure for ground truth establishment, 214 codings remained, as frames without consensus were excluded from analysis.

A. Preliminary Results

We conducted evaluations of the pipeline using off-the-shelf models (GLM-4.5V & InternVL-3.5 as open-sourced, Gemini-2.5-Pro & GPT-5 as commercial) to assess pipeline feasibility, characterize model failure modes, and compare model outputs with human annotations.

TABLE II
EVALUATION RESULTS WITH CoT PROMPTING ACROSS DIFFERENT VLMS. SESSIONS A (N=173) AND B (N=41) SHOW MEAN AND RANGE (MIN-MAX) ACROSS 5 RUNS; OVERALL (A+B) METRICS ARE COMPUTED BY POOLING ALL PREDICTIONS ACROSS SESSIONS AND RUNS (N=1070 TOTAL PREDICTIONS FROM 5 RUNS $\times$ 214 CODINGS).

Session	Metric	GEMINI-2.5-PRO	GLM-4.5V	GPT-5	INTERNVL-3.5
A	F1	0.794 [0.781, 0.804]	0.794 [0.770, 0.816]	0.799 [0.786, 0.819]	0.774 [0.753, 0.802]
	Acc	0.794 [0.780, 0.803]	0.794 [0.769, 0.815]	0.784 [0.775, 0.803]	0.770 [0.751, 0.798]
B	F1	0.502 [0.451, 0.530]	0.723 [0.673, 0.745]	0.537 [0.518, 0.571]	0.548 [0.497, 0.599]
	Acc	0.439 [0.390, 0.463]	0.742 [0.707, 0.756]	0.459 [0.439, 0.488]	0.537 [0.488, 0.585]
Overall	F1	0.744	0.783	0.754	0.730
	Acc	0.726	0.784	0.722	0.725

TABLE III
BASELINE EVALUATION RESULTS WITH PREDICTION-LEVEL AGGREGATION (N=1070). TWO CONDITIONS TESTED: “No-CoT” WITH RENDERED PRE-PROCESSING (EYE GAZE & APRILTAG OVERLAYS), AND “No-CoT + No-Preproc” WITHOUT VISUAL OVERLAYS.

Baseline	Metric	GEMINI-2.5-PRO	GLM-4.5V	GPT-5	INTERNVL-3.5
No-CoT	F1	0.751	0.790	0.745	0.733
	Acc	0.752	0.788	0.751	0.713
No-CoT + No-Preproc	F1	0.720	0.777	0.703	0.705
	Acc	0.729	0.785	0.710	0.695

Table II reports the evaluation results of different VLMS with chain-of-thought prompting across two experimental sessions (A and B). For each session, we ran each model five times and reported the mean performance, along with the observed range (minimum–maximum) across runs. We used F1-score and accuracy as the primary evaluation metrics. The **Overall** row aggregates all runs across both sessions and presents prediction-level mean results. As highlighted in bold, GLM-4.5V consistently outperforms other models in

"GLM-4.5v: Let me analyze this image carefully to identify the participants and classify their actions. From the image, I can see two people sitting at a table with various electronic components and a laptop. **Let me identify them based on the given participant descriptions:** Participant 6: Male with a black long-sleeve shirt/jacket and a white mask. Participant 10: Male with dark blue sweater. Looking at the image: - The person on the left is wearing a black shirt and has a white mask, which matches participant 6. - The person on the right is wearing a dark blue sweater, which matches participant 10. I can also see AprilTags on both individuals: - The person on the left (participant 6) appears to have an AprilTag. - The person on the right (participant 10) has a clear AprilTag visible on their chest. **Now let me extract the observational details for each participant:** For participant 6 (left): - **Gaze:** Looking down at a book/magazine they're holding. - **Hands:** Holding the book with both hands. - **Position:** Sitting at the left side of the table. - **Clothing:** Black long-sleeve shirt and white mask (matches participant 6 description). For participant 10 (right): - **Gaze:** Looking at the laptop screen (green gaze line shows this). - **Hands:** Both hands appear to be on or near the laptop keyboard. - **Position:** Sitting at the right side of the table. - **Clothing:** Dark blue sweater (matches participant 10 description). **Now I need to classify their actions using the decision process:** For participant 6: - Gaze is directed at a book (work-related element). - Hands are holding the book (contacting the gaze target). This would suggest "Manipulating" since gaze is at a work element and hands are contacting that same element. For participant 10: - Gaze is directed at the laptop (work-related element). - Hands appear to be on/near the laptop keyboard (contacting the gaze target). This would also suggest "Manipulating" since gaze is at a work element and hands are contacting that same element. Wait, let me double-check the gaze and hand positions more carefully. For participant 6: - Gaze: Looking down at the book they're holding. - Hands: Both hands are holding the book. - This does fit "Manipulating" as they're interacting with the book. For participant 10: - Gaze: The green line shows gaze directed at the laptop screen. - Hands: Both hands appear to be positioned over the laptop keyboard. - This also fits "Manipulating" as they're interacting with the laptop. However, let me reconsider the action definitions more carefully: "Manipulating": (1) GAZE is directed at a work-related element. AND (2) At least one HAND is clearly contacting the same GAZE target. Both participants seem to fit this definition. Participant 6 is manipulating the book, and participant 10 is manipulating the laptop. But wait, let me look at the image again more carefully. For participant 10, I can see the green gaze line going from their face to the laptop, and their hands do appear to be on or near the keyboard. This definitely looks like manipulation. For participant 6, they're holding a book and looking at it, which also fits manipulation. Actually, let me check if there are any other possibilities. Could participant 10 be "Observing" instead? No, because their hands appear to be actively interacting with the laptop, not just hovering or inactive. Could participant 6 be "Observing"? No, because they're actively holding and looking at the book, which is manipulation. So my final classification would be: - Participant 6: Manipulating (book). - Participant 10: Manipulating (laptop). **Let me structure this into the required JSON format.**"

Fig. 3. Reasoning content of frame 1 shown in Fig. 2 from GLM-4.5V with CoT prompting: Grounding (purple) - Captioning (red) - Classifying (green) - Formulating (blue)

Session B and in the overall average, whereas GPT-5 shows the strongest performance in Session A. Results also show that all models perform considerably worse in Session B, which contains more occlusions and thus represents a more challenging setting. Finally, the reported ranges demonstrate the stability of each model across repeated runs, indicating its reliability and reproducibility.

We found that explicit CoT prompting do not always improve performance; in fact, some models even showed a slight decline compared to their baseline without CoT prompts (Table III). We attribute this to the fact that many contemporary VLMs already possess implicit self-CoT capabilities, see ² and ³; even without explicit prompting, they still generate intermediate reasoning steps internally. We also tested a stricter baseline without both CoT prompting and rendered pre-processing (eye gaze and AprilTag). This "No-CoT + No-Preproc" setting yielded consistently lower performance (Table III), confirming the value of these perceptual cues. Nevertheless, explicit CoT prompting remains valuable for a different reason: it makes the model's intermediate observations and justifications transparent, as shown by Fig. 2. More importantly, revealing such reasoning traces from open-source models (Fig. 3) provides meaningful insights into model behavior and error patterns, even if raw performance gains are not guaranteed.

B. Class-wise analysis

Fig. 4 shows confusion matrices for the optimal runs of each VLM per session. Across models, **Manipulating** and **Observing** are most reliably detected, reflecting their clearer visual signatures. In contrast, **Communicating** is often misclassified, suggesting that non-verbal communicative signal

²https://platform.openai.com/docs/guides/reasoning-best-practices?utm_source=chatgpt.com

³https://ai.google.dev/gemini-api/docs/thinking?utm_source=chatgpt.com

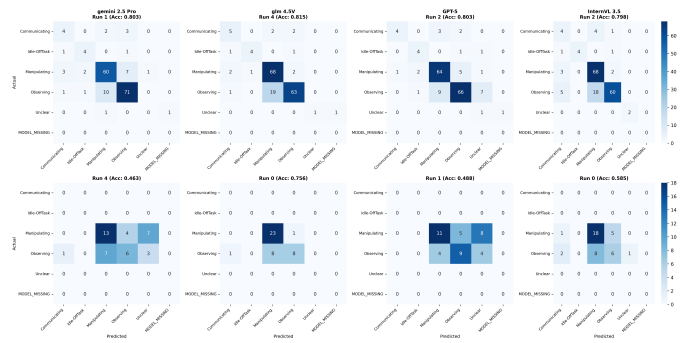


Fig. 4. Confusion matrices of VLMs for the optimal run. Top row: Session A; Bottom row: Session B

(e.g., gaze, gesture, body alignment) are difficult to infer reliably from single-frame visual input. The **Idle-OffTask** and **Unclear** categories occur less frequently. The additional category **MODEL_MISSING** denotes coding where a participant was present in the ground truth but was outside the camera view or otherwise not visually identifiable by the model.

V. DISCUSSION

This paper investigated two research questions: the first concerned the extent to which a VLM-based pipeline, augmented with gaze data, can automate collaborative action recognition with human-interpretable reasoning. The pipeline combines the identification of the individual, gaze direction, and hands, and then classifies actions based on CoT to determine the actions. There are advantages to automating the analysis of high-level human behavioural interactions for educational research, but ensuring that these systems are accurate, transparent, and explainable is a key concern [12], [20]. The results from the pilot test demonstrate good accuracy for the different VLMs, with the open-source model GLM [24] performing the best across both samples. An additional advantage of using open-source models is the ability to see the reasoning behind how these models followed the rule-based prompts to code the video frame (see Figure 3). Therefore, we can say that the pipeline integrated with AprilTag and gaze detection, combined with the Zero-Shot-CoT prompting, has the potential to recognize different actions in a collaborative learning activities context. More research across diverse datasets and models is needed to determine the best approach to use and how explicit the CoT prompts should be to maintain transparency.

Secondly, we explored the differences and similarities between the CoT AI pipeline and human perception of the same video frames. For most frames, there was good agreement between human coders and the VLM. However, in many instances the VLMs reported uncertainty when hands were occluded, while human coders inferred contact from posture and prior context. We argue that it is not a matter of whether humans or VLMs are right, since there are multiple ways to reconcile their views—prompts can be adjusted, coding rules renegotiated, or camera setups improved. The difference between machine and person needs to be viewed in terms of what

is required to answer the research questions that prompted the observations [9], [17]; the appropriate level of inference depends on the learning constructs under investigation [20].

This study is framed as a technical feasibility study rather than a statistical benchmark. The pilot dataset (N=214 person-frame codings) is limited but sufficient to validate whether VLM-based CoT can produce logically structured reasoning aligned with human annotation practice. The uneven class distribution (e.g., Communicating $n=9$, Idle-OffTask $n=6$) reflects the ecological reality of collaborative learning, where certain actions naturally dominate; forcing artificial balance would compromise ecological validity. Frames without annotator consensus were excluded to establish a clear gold standard for initial pipeline validation; for future deployment, we envision a human-in-the-loop workflow where the pipeline's CoT reasoning traces enable confidence-based triage, flagging uncertain frames for human review rather than forcing classification.

A. Limitations & Future research

The pilot study has limitations due to the two small datasets; however, the feasibility of using this pipeline on a larger scale, combined with audio analysis and the inclusion of multiple camera angles, can mitigate some of the occlusion issues and provide clearer data for the model reasoning. Additionally, the development of integrated tools for human coding and machine reasoning can provide a scalable pipeline that is hopefully reliable and transparent for research and practice using learning analytics. Future work will also report processing time statistics relevant for evaluating real-time deployment capabilities, and explore strategies such as synthetic data augmentation or active learning to address class imbalance for rare action categories.

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